

# Eco-Sort Smart Waste Management System

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**Abstract**—Population density in cities has caused a significant increase in waste production in urban areas in Indonesia, one of which is in cities like Bandung, which has challenges such as unclear information on Temporary Disposal Site (TPS) capacity, uncertain waste collection schedules, and low community participation. This paper discusses solutions by developing the EcoSort application, an application aimed at intelligent waste management that integrates real-time TPS navigation, fixed collection scheduling, a recycling material market (Eco Market), and an EcoPoints reward system into one platform. The development of this application uses the Agile Software Development Life Cycle (SDLC) method as an Android application using Java. EcoSort aims to present an effective, participatory, and economically profitable waste management application. This application can be a solution in waste management efficiency and encourage sustainable behavior among the community.

**Keywords**—smart waste management, mobile application, Agile SDLC, reward system, Android, EcoSort-

## I. INTRODUCTION

The rapid growth of urban populations and the rapid increase in economic development in many developing countries generate wastes that require efficient municipal solid waste (MSW) management systems. Planning that cities must prepare to face greater challenges, especially in terms of waste, must also be supported by efforts to improve sustainability systems for the environment, especially waste collection systems, and the quality of public services at affordable costs by local governments, and requires community participation in the decision-making process. However, despite a number of obstacles such as unpredictable waste collection schedules and difficulties in informing city residents about their household waste production patterns that cannot all be overcome, this integrated smart waste management is expected to have potential benefits related to reducing illegal dumping rates (85%), as well as optimizing logistics operations [1] and increasing community involvement in improving a better environment.

A good way to address this long-standing challenge is through the creation of a well developed digital infrastructure supported by mobile technology. This infrastructure is

designed to reduce information redundancy, improve the economy, and hopefully strengthen community engagement in waste management activities. In this paper, we focus on creating a waste management application to improve effective and user friendly waste collection services, supporting operational effectiveness and its impact on the surrounding environment. To create a more desirable smart waste management platform, the services provided need to be thoroughly analyzed and evaluated so that the system can be used easily and effectively. Structured Agile-based development techniques provide good evaluation results for performance and user engagement, as this is crucial for exploring a suitable waste management ecosystem.

This approach is centered on a continuous iterative pattern from initial design to design implementation from user feedback based on actual interests. The EcoSort mobile application was developed for urban waste management in this study. It is hoped that this application can be a solution to the problem of household waste or garbage that is integrated with TPS navigation, definite collection scheduling, as well as a recycling market to support the management of second-hand goods, and a reward-based incentive system, because this system has not been fully implemented in the existing system in Indonesia today. Because until now, most waste management applications and systems still use basic features such as monitoring trash bin capacity, simple tracking in waste collection and very weak user engagement mechanisms [2][3].

In today's technological age, an ideal waste management service relies entirely on the use of mobile technology. Navigation features to disposal sites, real-time data, and reward-based systems to motivate people to do proper sorting during waste collection are lacking. Thus, the integration of designed features is expected to increase user engagement in most cities, especially in Indonesia [4]. Related research has focused on smart waste management systems in monitoring the condition of waste bins [1], basic application features in scheduling waste collection times [5] and market concepts aimed at the recycling market of recyclable goods [6].

Some studies performed more detailed assessments of mobile-based waste management applications [7][8], analyzing various features and adoption rates to determine how these

tools have influenced community behavior. All successful Urban Waste Management programs in any part of the world will face geographical, social and logistic challenges [9]; Therefore it is very important to design and predict system characteristics according to local context conditions such as for example Bandung (West Java) [3]. Although many existing findings appear consistent with IoT-based or partial digital solutions that may be satisfying, they generally lack consistency and more compare unfavorably to a holistic mobile platform where navigation transparency, scheduling security economic incentives exist together in combination (implicitly) along with user reward systems to drive high levels of participation by a given community [2][10].

A very challenging area for the researchers working in this domain is to formulate a high extensible smart waste management application that deals with both technical and behavioral aspects related to urban areas of developing countries. Waste management in high-density cities has specific features which means that it cannot be treated as conventional systems but requires a different and integrated approach. Thereby, this paper focuses on the design and implementation of EcoSort mobile application for smart waste management. The implementation of the Agile Software Development Life Cycle (SDLC) methodology outlined in sections that follow serves as an example for using SDLC methodologies to realize a system architecture and features designed into parallel development through iterative refinement.

The remainder of this paper is organised as follows: Section II develops the problem formulation used for realistic urban conditions. An overview of our proposed EcoSort system and its basic features and architecture is detailed in Section III. Section IV describes the Agile SDLC adopted for use during its development. Section V describes the implementation results and system demonstration. And last, Section VI concludes with future work.

## II. LITERATURE REVIEW

Urbanization is the main driver of rapid increase in municipal solid waste (MSW) generation, compelling massive research towards smart waste management solutions. In the last 10 years, multiple technological solutions have been suggested to improve waste collection, monitoring and segregation, and incentivising communities. This section presents a literature review related to the development of the EcoSort application.

IoT is the common theme of several studies for smart waste management. Gopi *et al.* proposed an IoT based waste management system that detects the filled level of the waste bins using ultrasonic sensors and provides feedback to the authorities using a notifier so that the waste can be collected in time [1]. This has been a tried and tested mechanism to lower operational costs and optimise routes for collection. Likewise, Sosunova and Porras [2] performed an exhaustive systematic review of dozens of IoT-enabled smart waste management systems implemented in smart cities. Their paper concluded that IoT solutions are very effective for improving waste collection efficiency. However, most of the applications focus primarily on backend operations (authorities and waste collectors) and do almost nothing for obtaining direct feedback from the end-users (the citizens). This creates a significant gap between technological advancement and actual community participation.

Most mobile applications currently developed in the field of circular waste management focus their exploration on improving service accessibility and user engagement. Two relevant publicly published studies have been released: Tundjungsari *et al.* [5] proposed an integrated mobile application model that supports the full waste value chain; Rimbawa *et al.* [6] designed an application system that integrates the Google Maps API and QR codes, which can support organic waste collection and recycling transactions between merchants and consumers. These platforms can narrow the gap between the general public and waste management services, but most existing solutions have fragmented functions, only able to provide basic features such as tracking, information push, and scheduling, and have not integrated multiple types of core functions.

Multiple local studies conducted in Bandung, Indonesia have all confirmed prominent problems in the city's urban waste management. Rahmasary *et al.* [3] assessed municipal governance issues in Bandung related to waste, water, and climate change, and proposed countermeasures to advance digital transformation and strengthen citizen participation; Siswantini *et al.* [9] pointed out that insufficient waste management literacy hinders the development of sustainable cities. New Security Beat [7] and WWF [8] documented two existing types of digital waste initiatives in Indonesia: applications that serve waste collectors, and the digitalization of Bank Sampah (waste banks). While these initiatives hold the potential to generate greater economic value, they suffer from core flaws: they are mutually isolated, lack real-time functions, and have weak, cumbersome incentive mechanisms. These conditions lay a reasonable foundation for the implementation of the present study.

In existing research on urban waste governance, boosting public participation through reward and gamification measures is a core subfield. Numerous external studies have confirmed that incentive systems can significantly encourage users to sort their waste and dispose of it in compliance with relevant rules. Only a very small number of studies have integrated reward systems with the route navigation and collection scheduling functions of TPS (temporary waste processing sites), and this combination of functions is particularly critical for the complex waste collection and transportation logistics scenarios in developing countries.

From an expert perspective on software development, Mathur *et al.* [10] conducted an analysis comparing various methodologies in mobile application development. In conducting software development, Rahimian *et al.* [11] focused on identifying suitable Agile methodologies specifically designed for mobile software projects. They found and confirmed that Agile SDLC offers superior flexibility, with faster iteration processes, and better adaptability to changing demands from user feedback compared to traditional approaches. Based on these characteristics, the Agile method is very suitable for academic projects and projects carried out by students with time constraints and needs and interests that will continue to evolve.

Existing research in the field of smart waste management has accumulated a certain practical and theoretical foundation, yet it still faces three core limitations: IoT systems focused on hardware monitoring suffer from the shortcoming of limited single-function design[1], mainstream mobile applications only cover a single domain of the full waste lifecycle[5], and local Indonesian projects generally have the problems of low technology integration and informal application scenarios[3][8]. At present, the field lacks a one-stop, user-friendly mobile application that adapts to the

sociotechnical context of cities such as Bandung and integrates four a comprehensive smart waste management mobile application termed **EcoSort**. This project aims to use the Agile Software Development Life Cycle (SDLC) methodology

[10][11] to provide a more cohesive solution that is better aligned with wide application and practical needs compared to existing literature.

### III. METHODOLOGY

The Agile Software Development Life Cycle (SDLC) methodology was followed in this project for the development of the EcoSort smart waste management mobile application. The Agile methodology was selected for its iterative, flexible and collaborative approach, particularly suited for mobile app development with changing requirements and short development time.

#### A. Overview of Agile SDLC

Agile SDLC is an incremental and iterative process that emphasises functional software delivery through short development cycles, known as sprints. Agile is different from traditional Waterfall model in a way that it allows continuous feedback, changes and close collaboration between team members. [10][11].

The Agile process implemented in this project is illustrated in the following diagram:

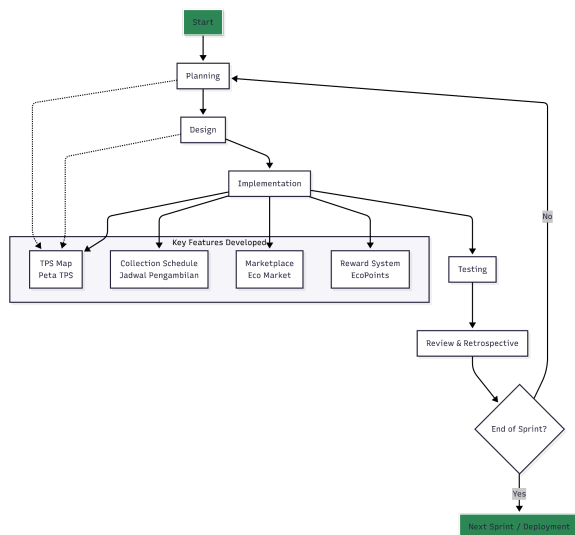


Fig.1. Agile SDLC workflow in EcoSort Development

#### B. Reasons for Choosing Agile Methodology

There are many good reasons to choose the Agile methodology that fits the needs of this project:

1. **Parallel Development Effectiveness** With Agile, the team breaks the work down into small tasks, so different parts can be developed at the same time. For example, UI/UX design, XML layout design, Intent logic, API integration (Google Maps) can all be done simultaneously without any dependency bottlenecks.
2. **Strong, Team Collaboration** The methodology allows team members to work simultaneously. The interface development in Android Studio, functional coding, database designing and documentation were done in parallel. Extensive utilisation of GitHub version control to manage code changes and facilitate smooth collaboration.
3. **Feedback and Continuous Improvement** Agile strongly advocates continuous evaluation. Feedback can be given during sprint reviews by lecturers, peers

or potential users and can be directly implemented in the next iteration. This iterative process was instrumental in shaping the user interface, feature flow and overall user experience of the EcoSort application.

#### C. Tools and Technologies Used

EcoSort was built with the following tools and technologies:

- developed mobile application EcoSort includes 6 tools,
- such as Android Studio,
- Kotlin, and Google Maps API. Each tool is
- assigned a unique, project-specific function, and together
- they comprehensively cover all core
- requirements across the full scope of full-link development.
- 

#### D. Implementation of Agile in EcoSort Development

The development process was divided into several sprints Each sprint had a structured cycle of:

- sprint is set with a standardized work
- cycle, and work proceeds sequentially
- through sprint planning, prototype design, coding development,
- multi-type testing, and review and retrospection.
- Tasks for all work links are clearly defined, with no blind spots in execution.

This project carries out development in a small-scale, phased manner, centered on four core functions: TPS Map, Collection Schedule, Marketplace, and the EcoPoints Reward System. A minimum viable product (MVP) is delivered in the early stage of the development cycle, after which continuous iterations are

completed to integrate all full functionalities. The project adopts the agile SDLC methodology, which perfectly meets the complex development requirements of integrating multiple categories of functions, maintaining a coherent user experience, and strictly adhering to set time limits.

### IV. PROPOSED SYSTEM

EcoSort, deployed on the Android platform, is a mobile tool for urban waste management. Centered on citizens' needs, it builds a closed management loop that enables public participation by relying on three categories of supporting mechanisms, and achieves full-process digital optimization of the entire work process.

#### A. System Architecture

EcoSort adopts a client-server architecture with three main layers:

1. **Presentation Layer (User Interface)** — Built using Android XML layouts and Kotlin.
2. **Business Logic Layer** — Handles core functionalities such as mapping, scheduling, marketplace

transactions, and reward calculations.

3. Data Layer — Utilizes Firebase for cloud database and authentication, combined with SQLite for offline capabilities.

Google Maps API and FirebaseThe app heavily relies on Google Maps API for all geographical aspects, including delivery availability and user geolocation, while Firebase provides real-time data synchronization for the service.

## B. Key Features of EcoSort

The system is composed of 4 major modules and some supporting features that are:

- Peta TPS (TPS Navigation) Pengguna dapat melihat lokasi TPS terdekat lengkap dengan jarak, jam operasional, dan status aktif/not aktif. The feature supports route navigation and search
- Jadwal Pengambilan Sampah (Collection Schedule) This module gives predictive schedules for waste collection from municipal and private transporters, respectively. Avoiding the accumulation of waste with the notifications received by the users during with enough time periods before the scheduled time of collection.
- Eco Market (Marketplace for Recyclables) A specially designed product listing platform for users to sell recyclable goods (plastic, paper, cardboard, metal, etc) or users can post ads (ads in English) Transactions can be made with EcoPoints or cash.
- EcoPoints Reward System: Users can earn by disposing of waste in TPS, selling recyclables, or completing eco-challenges. This points can be exchanged to many other merchants in exchange for e-wallet balance, electricity, phone credit, merchandise, and so on.

Additional features include:

- User profile and authentication
- Transaction history
- Customer service (WhatsApp chat and email support)

## V. EXPERIMENTAL RESULTS

The EcoSort app was developed completely on Android Studio using Kotlin primarily. The development used the Agile SDLC as outlined in Section III.

### A. Implementation Details

- Development Environment: Android Studio Minimum SDK : 24 (Android 7.0)
- Major Libraries: Google Maps SDK, Firebase Authentication, Firestore database and Storage library from firebase (Image loading with Glide).
- DESIGN: Material Design 3 (MD3) ~ Dark Theme → contemporary and easy on the eyes
- Offline Support: Using SQLite you can still read important information such as list of saved TPS locations and user profiles.

### B. Main Features Implementation

1. TPS Peta digunakan untuk menampilkan peta interaktif menggunakan Google Maps API. You can

filter by type and TPS, in addition to real-time navigation.

2. Jadwal Pengambilan Sampah Menampilkan jadwal pengambilan dengan status warna (On Time, Delayed dan Completed) Firebase Cloud Messaging (FCM) is used for push notifications.
3. Eco Market Users have the capability to upload pictures of recyclables and determine prices either in points or cash, as well as manage buy/sell transactions.
4. Reward System: Points are automatic and added to users when they report successful waste disposal or conduct a transaction. There are some partner vouchers included in the redemption system.
5. Supporting Features
  - Secure user login and registration
  - Complete transaction and reward history
  - 24/7 customer service integration

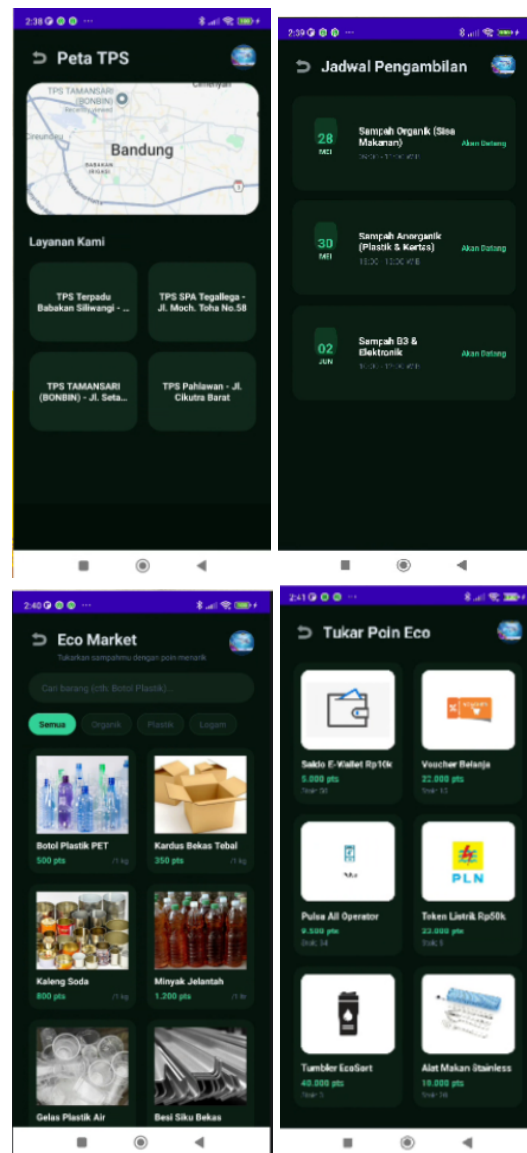


Fig. 2. Screenshot of EcoSort Main Features (Peta TPS, Jadwal Pengambilan, Eco Market dan Sistem Reward)

### C. Testing and Evaluation

This application was tested using unit testing, integration testing and user acceptance test with a few further respondents. It was a test to primarily check functionality, usability and performance across multiple Android devices. Overall, reviewers praised the app for its ease in use and usefulness of

features.

The provided result is an evidence of a clean-built down-to-earth simple smart waste management application evolved in short timeframes through Agile platform-based usability utilizing mobile technologies.

## VI. CONCLUSION

### A. Implication

Since the country's population and urbanization have caused a tremendous growth in the annual creation of municipal solid garbage, the study tackles the waste management issue in Indonesia. The traditional waste management is still inefficient since garbage collection is not constant, TPS does not have real time information and the community does not care about sorting the waste. All this means recycling is rare and illegal dumping is ubiquitous.

The purpose of this project was to address these problems by developing EcoSort, a whole mobile application that integrates a digital solution for smart garbage management.

### B. Overview of the Suggested Solution

EcoSort is a mobile application that was developed for Android devices. It is a software solution which combines several basic functions in a single program. A Google Maps API is available for real-time TPS navigation, there's an on-demand autonomous waste collection system, Eco Market is a marketplace for recyclable products, and users may earn EcoPoints to reward their friends. Additionally, it included the addition of additional features, such as the capability to command user profiles, view a history of transactions, and get in touch with customer service. Agile Software Development Lifecycle (SDLC) The Agile software development life cycle approach was used to make the whole app so that it could be flexible and keep getting better.

### C. Analysis of Development Process

Our team adopted agile development technologies to build a mobile waste management application, simultaneously advanced development of the application's ecological module, established a multi-cycle closed feedback loop, and ultimately delivered a qualified, user-friendly application that covers both technical and behavioral dimensions.

### D. Comparison with Previous Studies

The EcoSort intelligent waste management platform developed in this paper aligns with early research in the field. Most existing comparable platforms only rely on the Internet of Things (IoT) to carry out single tasks such as cross-city waste tracking and garbage bin monitoring, and such integrated functions are extremely scarce in Indonesia, particularly in Bandung.

### E. Practical Implications

EcoSort, the waste management software independently launched by our team, can effectively improve waste management efficiency and boost community participation. The data compression technology built into the software supports core governance goals including curbing illegal dumping and optimizing waste collection and transportation routes. The software is also compatible with most low-end Android devices, allowing it to cover the vast majority of

ordinary residents.

### F. Conclusion

This study reaches core conclusion that EcoSort, the smart waste management application developed using agile software engineering, can sustain citizen participation through real-time data and economic incentives after its deployment in Indonesian cities, and advance sustainable waste disposal, community action, and the transition to a circular economy.

### G. Recommendations

After our team conducted on-site inspections of the operation of EcoSort, a local waste sorting application in Bandung, Indonesia, we propose that for the application to truly achieve functional on-the-ground implementation, several core adjustments that precisely target its current deployment bottlenecks must be rolled out: As the highest-priority step, coordinate with Bandung's municipal waste department to resolve the unaligned vacuum operation issue caused by a lack of synchronized public data and official endorsement; collaborate with the local Bank Sampah network, private recycling enterprises, and established e-wallet platforms to grant the EcoPoints earned by users real practical value; host hands-on workshops and grassroots training sessions in local communities to overcome the technical barrier that leaves residents unable to use the application; and require the development team to maintain product flexibility, continuously pushing out small updates based on real user feedback to fix performance bugs.

### H. Limitations

The waste management pilot program of this study has achieved preliminary outcomes, yet it still has limitations: it focuses solely on the ordinary citizen end, has a low level of system integration with official bodies and waste collection and transportation operators, has only completed small-scale testing, and its long-term impacts have not been comprehensively assessed.

### I. Future Scope

The EcoSort smart digital waste management scheme proposed in this paper has five planned R&D expansion directions for future development: equipping smart trash bins with IoT sensors to realize real-time monitoring, using machine learning to optimize waste collection and transportation routes, expanding deployment from Bandung, Indonesia to multiple cities, building an analysis module to generate waste generation trend reports for local governments, and adopting blockchain technology

to ensure the transparency of reward transactions. This easy-to-use, practical digital solution represents a reliable advancement toward AI-powered, community-oriented sustainable waste management, and can help Indonesia address its urban waste problems.

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